

Assignment #4 Report

Customer Churn Prediction with XGBoost

Dataset: IBM Telco Customer Churn | Submission: Spring 2026

Part 1: Data Exploration

1.1 Dataset Overview

The IBM Telco Customer Churn dataset contains records for 7,043 customers of a telecommunications company. Each row represents one customer and includes demographic information, details about subscribed services, billing information, and whether the customer ultimately churned.

Property	Value
Rows (customers)	7,043
Columns (features + target)	21
Target variable	Churn (Yes / No)
Churn rate	26.5% (1,869 Yes / 5,174 No)
Missing values (raw)	0 explicit NaN: 11 hidden blanks in TotalCharges

1.2 Feature List

The 20 predictor features (after dropping customerID) are:

Feature	Type	Description
gender	Categorical	Male / Female
SeniorCitizen	Numeric (0/1)	Whether the customer is a senior citizen
Partner	Categorical	Whether the customer has a partner (Yes/No)
Dependents	Categorical	Whether the customer has dependents (Yes/No)
tenure	Numeric	Number of months with the company
PhoneService	Categorical	Whether phone service is subscribed
MultipleLines	Categorical	Whether multiple phone lines are active
InternetService	Categorical	DSL / Fiber optic / No

OnlineSecurity	Categorical	Online security add-on (Yes/No/No internet)
OnlineBackup	Categorical	Online backup add-on
DeviceProtection	Categorical	Device protection add-on
TechSupport	Categorical	Tech support add-on
StreamingTV	Categorical	Streaming TV add-on
StreamingMovies	Categorical	Streaming movies add-on
Contract	Categorical	Month-to-month / One year / Two year
PaperlessBilling	Categorical	Paperless billing (Yes/No)
PaymentMethod	Categorical	Electronic check / Mailed check / Auto bank/CC
MonthlyCharges	Numeric	Current monthly charge in USD
TotalCharges	Numeric (string in raw)	Total amount charged to date

1.3 Problem Type & Business Context

Problem type: Binary classification: predict whether a customer will churn (Yes=1) or not (No=0).

Business importance: Customer churn is one of the most costly challenges in subscription-based businesses. Research consistently shows that acquiring a new customer costs 5 to 10x more than retaining an existing one. With 26.5% of customers churning, even a modest improvement in prediction accuracy can save millions in a real company. A churn model allows a telecom to proactively offer retention incentives to high-risk customers before they leave, reducing revenue loss and improving long-term customer lifetime value.

Part 2: Data Preprocessing

2.1 Cleaning Steps

Drop customerID:	Pure identifier column with no predictive value. Including it could cause overfitting if the model memorizes individual IDs.
Encode target:	Map Churn Yes → 1, No → 0. Required for sklearn and XGBoost which expect numeric labels.
Fix TotalCharges:	The column is stored as a string in the raw CSV. 11 rows contain blank strings (customers with 0 tenure who just signed up). <code>pd.to_numeric(..., errors='coerce')</code> converts these to NaN, which the median imputer handles downstream.

2.2 Feature Identification

Features were categorized explicitly rather than relying on dtype inference (which can be fragile across pandas versions):

Type	Features (4 numeric, 15 categorical)
Numeric	SeniorCitizen, tenure, MonthlyCharges, TotalCharges
Categorical	gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, De

2.3 Preprocessing Pipelines

Numeric pipeline: Median imputation (robust to outliers, handles the 11 NaN TotalCharges values) followed by StandardScaler (centers and scales features: essential for Logistic Regression, harmless for XGBoost).

Categorical pipeline: Most-frequent imputation (safe default; no categorical columns have missing values in this dataset, but ensures robustness to new data) followed by OneHotEncoder with handle_unknown='ignore' to avoid errors on unseen categories.

Train/test split: 80/20 split with stratify=y to preserve the 26.5% churn rate in both sets. Random state 42 for reproducibility.

2.4 Key Code Snippet

```
numeric_transformer = Pipeline(steps=[ ('imputer', SimpleImputer(strategy='median')),
('scaler', StandardScaler()) ]) categorical_transformer = Pipeline(steps=[ ('imputer',
SimpleImputer(strategy='most_frequent')), ('onehot', OneHotEncoder(handle_unknown='ignore',
sparse_output=False)) ]) preprocessor = ColumnTransformer(transformers=[ ('num',
numeric_transformer, numeric_features), ('cat', categorical_transformer,
categorical_features) ])
```

Part 3: Baseline Model: Logistic Regression

Logistic Regression serves as the baseline. It is a well-understood linear model that provides a strong reference point. If a more complex model cannot beat it, the additional complexity is not justified. max_iter=1000 ensures convergence on the encoded feature space (~45 columns after one-hot expansion).

3.1 Baseline Results

Metric	Score
Accuracy	0.8055
Precision	0.6572

Recall	0.5588
F1-score	0.6040
ROC-AUC	0.8419

3.2 Confusion Matrix (Baseline)

Of 1,409 test customers, the baseline correctly classified 926 non-churners and 209 churners. However it missed 165 actual churners (false negatives) : customers who left without being flagged for retention. This false negative count is the primary business concern.

Part 5: Evaluation Metric Choice

Chosen Metric	ROC-AUC (Area Under the ROC Curve)
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Why ROC-AUC?

ROC-AUC measures a model's ability to rank customers by churn probability across all possible decision thresholds. In a real deployment, the model generates a ranked list of at-risk customers and the retention team targets the top K%. ROC-AUC directly measures the quality of that ranking.

Why not Accuracy?

The dataset is class-imbalanced: 73.5% of customers do not churn. A trivial model that predicts nobody ever churns achieves 73.5% accuracy while being completely useless. ROC-AUC is immune to this trap because it evaluates both the positive and negative class performance jointly and does not depend on a single threshold.

Why ROC-AUC over F1?

F1-score is computed at the default 0.5 decision threshold and cannot be used to compare models at different operating points. ROC-AUC is threshold-independent, making it the right metric for model selection when the final threshold will be tuned separately based on the business cost of false positives vs. false negatives.

Part 4: XGBoost Model + Hyperparameter Tuning

4.1 Hyperparameters Tuned

Hyperparameter	Values Searched	Rationale
n_estimators	100, 200	Controls the number of boosting rounds. More trees can improve accuracy but risk o
max_depth	3, 5	Controls tree depth. Shallow (3) = conservative; deeper (5) = captures more comple
learning_rate	0.05, 0.10	Shrinks each tree's contribution. Lower rate + more trees typically generalizes bette
subsample	0.8, 1.0	Fraction of training rows used per tree. Values <1 add stochastic regularization, re

4.2 Tuning Method

GridSearchCV with 3-fold cross-validation was used to evaluate all 16 combinations (2⁴ grid). Scoring was set to roc_auc to match the primary evaluation metric. n_jobs=-1 parallelizes the search.

```
param_grid = { 'classifier__n_estimators': [100, 200], 'classifier__max_depth': [3, 5],
'classifier__learning_rate': [0.05, 0.1], 'classifier__subsample': [0.8, 1.0] } grid_search =
GridSearchCV( estimator=xgb_model, param_grid=param_grid, scoring='roc_auc', cv=3, n_jobs=-1,
verbose=1 ) grid_search.fit(X_train, y_train)
```

4.3 Best Hyperparameters

After grid search, the optimal configuration was:

Parameter	Best Value
n_estimators	200
max_depth	4
learning_rate	0.05
subsample	0.8
CV ROC-AUC	0.8461

4.4 XGBoost Test Results

Metric	Score
Accuracy	0.8148
Precision	0.6823
Recall	0.5775
F1-score	0.6256
ROC-AUC	0.8548

Part 6: Feature Importance & Interpretation

6.1 Top 10 Most Important Features

Feature importance was extracted from the trained XGBoost model using the default 'gain' metric, which measures the average information gain each feature contributes across all splits in all trees.

Rank	Feature	Importance	Business Meaning
1	Contract: Month-to-month	0.311	No lock-in = easy to leave
2	TotalCharges	0.159	Proxy for tenure & loyalty
3	tenure	0.121	Length of customer relationship
4	MonthlyCharges	0.119	High bills drive dissatisfaction
5	InternetService: Fiber	0.076	Premium product, higher expectations
6	OnlineSecurity: No	0.041	Fewer add-ons = less sticky
7	TechSupport: No	0.034	No support = frustration & churn
8	PaymentMethod: E-check	0.028	Manual payers re-evaluate monthly
9	MultipleLines: No	0.012	Lower service depth
10	PaperlessBilling: Yes	0.010	Digital customers are more mobile

6.2 Plain-Language Interpretation

1. Contract Type (Month-to-month): 31.1% importance

This is overwhelmingly the most important predictor. Customers on a month-to-month contract have zero financial penalty for leaving and can cancel at any time. Customers on annual or two-year contracts are locked in and rarely churn mid-term. **Business action:** Prioritize converting month-to-month customers to term contracts through incentives like discounts or service upgrades.

2. TotalCharges: 15.9% importance

Total charges reflect how much a customer has spent over their entire relationship. Low total charges typically indicate a new customer who has not yet built loyalty. These customers are the most volatile: they are still evaluating the service and have not invested enough in it to make switching costly.

3. Tenure: 12.1% importance

Customers who have been with the company longer are far less likely to churn. They have ported numbers, set up autopay, and integrated the service into their routines. Switching costs accumulate over time. **Business action:** Invest especially heavily in retention during a customer's first 12 months, when churn risk is highest.

4. MonthlyCharges: 11.9% importance

Higher monthly bills are associated with higher churn probability. Customers who feel they are overpaying: especially fiber optic subscribers: are more likely to compare competitors. **Business action:** Audit whether high-charge customers receive value proportional to their bill.

5. Fiber Optic Internet: 7.6% importance

Fiber optic customers churn at notably higher rates despite paying a premium. This likely reflects unmet expectations: customers paying for a top-tier product expect top-tier reliability and support. If service quality does not match the price, dissatisfaction escalates quickly.

6 to 7. No Online Security / No Tech Support: 4.1% + 3.4%

Customers without these add-ons are more likely to churn. Add-on services deepen the customer relationship and increase switching costs. Without them, the customer has fewer reasons to stay. **Business action:** Offer free trials of security and support add-ons to at-risk customers: stickiness increases adoption.

8. Electronic Check Payment: 2.8% importance

Customers who pay by electronic check (manually) churn more than those on automatic credit card or bank transfer. Auto-pay customers have one less decision point each billing cycle, reducing the monthly opportunity to reconsider the service. **Business action:** Incentivize switching to autopay.

Part 7: Final Comparison & Reflection

7.1 Side-by-Side Metric Comparison

Metric	Logistic Regression	XGBoost (Tuned)	Improvement
Accuracy	0.8055	0.8148	+0.0093
Precision	0.6572	0.6823	+0.0251
Recall	0.5588	0.5775	+0.0187
F1-score	0.6040	0.6256	+0.0216
ROC-AUC	0.8419	0.8548	+0.0129

7.2 Which Model Performed Better?

XGBoost outperforms Logistic Regression on all five metrics. The most meaningful gains are in **Precision (+2.5 pp)**, **F1-score (+2.2 pp)**, and **ROC-AUC (+1.3 pp)**. In absolute terms the improvements are moderate, but ROC-AUC rising from 0.8419 to 0.8548 means XGBoost produces meaningfully better-calibrated churn probability rankings: which directly translates to better targeting in a real retention campaign.

7.3 Why XGBoost May Work Better on This Dataset

Logistic Regression is a linear model: it can only learn relationships of the form 'as X increases, churn probability increases linearly'. The Telco Churn dataset contains non-linear relationships and important feature *interactions* that a linear model cannot capture without explicit manual engineering:

- The relationship between MonthlyCharges and churn is non-linear: customers at very low charges (new promos) and very high charges both show elevated churn, but for different reasons.
- Feature interactions matter: Contract=Month-to-month AND tenure < 12 AND Fiber optic is far more predictive than any single feature.
- XGBoost builds an ensemble of decision trees, each of which learns complex non-linear decision boundaries automatically.
- XGBoost also natively applies L1/L2 regularization, which reduces overfitting compared to unregularized Logistic Regression.

7.4 One Limitation of XGBoost

Limitation	Interpretability. Unlike Logistic Regression, where a coefficient directly tells you 'each additional month of tenure reduces log-odds of churn by X', XGBoost's ensemble of hundreds of trees cannot be read or explained to a non-technical stakeholder in plain terms. In regulated industries (banking, insurance, healthcare) or in any customer-facing application where teams must justify individual predictions, this is a real barrier. SHAP values can partially address this but add complexity to the workflow.
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Implementation Summary

Data Loading	Loaded 7,043-row CSV; identified TotalCharges as string type requiring conversion.
Preprocessing	Dropped customerID; binary-encoded Churn; coerced TotalCharges to numeric; built ColumnTransformer pipeline with median imputation + scaling for numeric, most-frequent + OneHotEncoder for categorical.
Baseline	Logistic Regression (max_iter=1000) via sklearn Pipeline. Achieved ROC-AUC = 0.8419.
XGBoost Tuning	GridSearchCV over 4 hyperparameters (16 combinations), 3-fold CV, scoring=roc_auc. Best config: n_estimators=200, max_depth=4, learning_rate=0.05, subsample=0.8.
Metric Choice	ROC-AUC chosen as primary metric: threshold-independent, robust to class imbalance, appropriate for ranked retention campaigns.
Feature Importance	Extracted via fitted_xgb.feature_importances_; top predictor is month-to-month contract (31.1%), followed by TotalCharges, tenure, MonthlyCharges, and fiber optic internet.
Model Selection	XGBoost outperforms baseline on all 5 metrics. Primary gain in ROC-AUC: 0.8419 → 0.8548 (+1.3 pp).